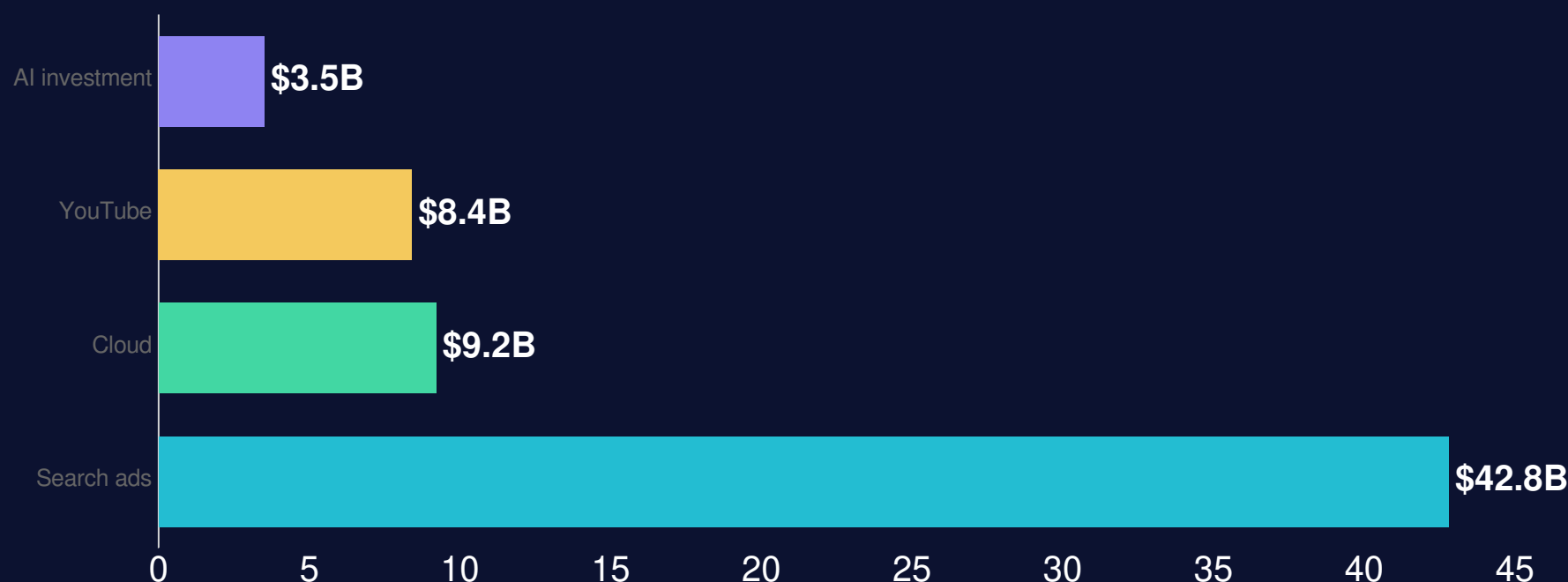


I built a prompt-to-chart AI prototype

A short business update becomes structured data, a chart, and a simple explanation.



Illustrative values from the notebook prompt

One paragraph, three useful outputs

The user writes normally. The pipeline handles the structure.

INPUT

Google Q2 2024
earnings:

Search ads: \$42.8B
Cloud revenue: \$9.2B
YouTube ads: \$8.4B
AI investment: \$3.5B

Plain text

01 Structured data

Names, values, periods, changes, and units in JSON.

02 Visualization

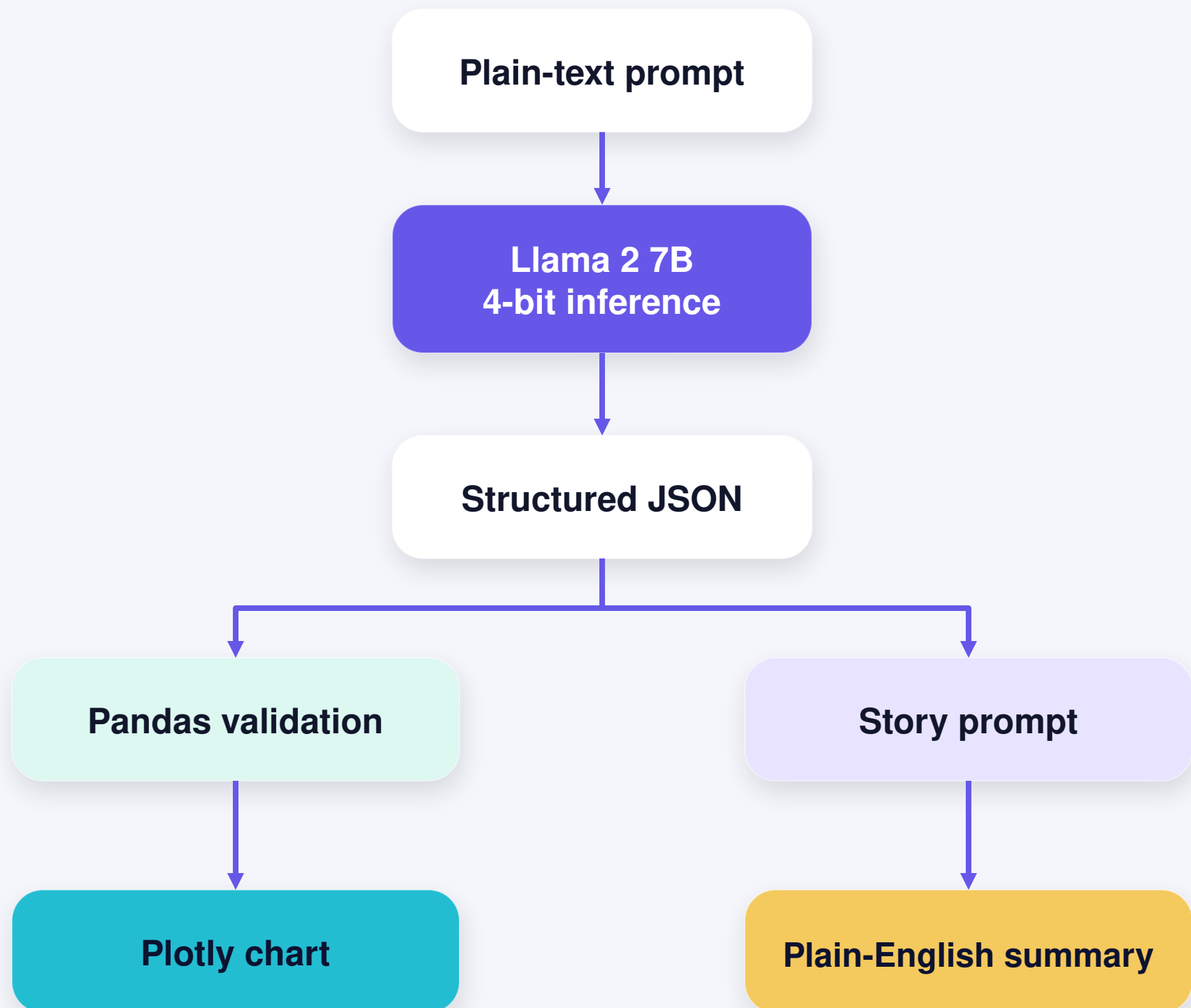
A clean Plotly chart built from the validated values.

03 Simple summary

A short explanation that uses the extracted facts.

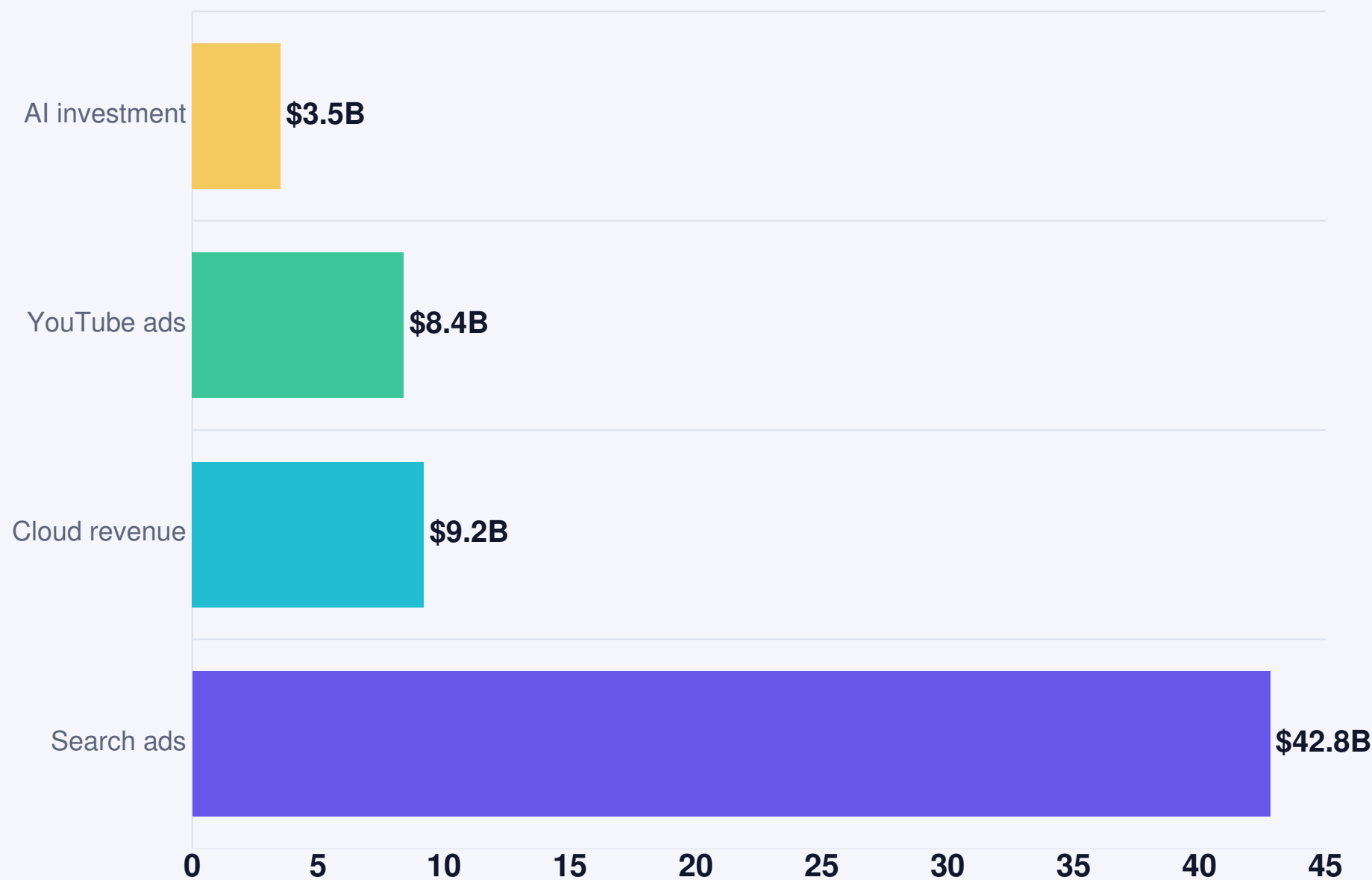
System architecture

The LLM handles language. Python checks the output and creates the chart.



Example output

The chart compares values that share the same unit.



Search ads is the largest value in this demonstration input.

Four lessons from the prototype

A good chart depends on more than a good prompt.

01 **Format control**

A fixed JSON schema gives the chart predictable input.

02 **Validation**

Code should check periods, magnitudes, missing values, and units.

03 **Comparable units**

Dollar values and percentages need separate chart axes.

04 **Human review**

The original source remains the final authority.

The next version

The notebook proves the basic flow. These additions would make it easier to use and test.

1

File input

Read CSV, Excel, and PDF files

2

Chart choice

Match the chart to the data shape

3

Simple interface

Add a Streamlit app

4

Evaluation

Measure extraction accuracy

Which input should I add first: CSV, Excel, or PDF?